RUQI SUN

+86 150 4413 6077 ◇ Shenzhen, China ◇ sunrq2024@mail.sustech.edu.cn

Website ◇ Google Scholar ◇ GitHub ◇ LinkedIn

EDUCATION

Southern University of Science and Technology Sep 2024 - Jul 2028
B.Eng. in Computer Science and Technology *Shenzhen, China*
GPA: 3.74/4.00 | **Advisor: Prof. Yuxin Ma**

University of California, Davis Jul 2026 - Aug 2026
Global Research Experience in Advanced Technologies (GREAT) *Davis, CA, USA*

National University of Singapore Jun 2025 - Jul 2025
NUS School of Computing Summer Workshop *Singapore*
Ranked 1st in the UI/UX Design cluster (A+) | Advisor: Prof. Bimlesh Wadhwa

RESEARCH INTERESTS

Human-AI Interaction · **Visualization** · **Creativity Support Tools**

PUBLICATIONS

[1] **Ruqi Sun**, Jiaping Li, Wenhui Tao, Ximing Zheng, Yuefeng Tan, Jiahao Wei, and Yuxin Ma†. "VeriForge: Mitigating Latent Knowledge Gaps in Narrative Drafting via Mixed-Initiative Scaffolding." Accepted to the 39th Annual ACM Symposium on User Interface Software and Technology (**UIST 2026**).

[2] Huanchen Wang, Qiuming Chen, Zhonghao Ji, **Ruqi Sun**, Zhichao Lu, and Yuxin Ma†. "SiMUSation: An Interactive Visitor Experience Simulation Framework to Support Museum Exhibition Design." Accepted to the 39th Annual ACM Symposium on User Interface Software and Technology (**UIST 2026**).

† Corresponding author.

RESEARCH EXPERIENCE

Research Intern, VIA Lab, University of California, Davis Jul 2026 – Present
Advisor: Prof. Dongyu Liu *Onsite Jul–Aug 2026; remote thereafter*

Undergraduate Researcher, DVA Lab, SUSTech 2025 – Present
Advisor: Prof. Yuxin Ma *Shenzhen, China*

SELECTED RESEARCH PROJECTS

Reflective Authorship in Visualization Authoring with Coding Agents July 2026 – Present
Project Lead, First Author, VIA Lab (UC Davis) & DVA Lab (SUSTech)
Advisors: Prof. Dongyu Liu and Prof. Yuxin Ma

- Led a formative study (N=12) of visualization authoring with coding agents, identifying ten recurring breakdowns across problem framing, data and analysis, and visual communication; derived a taxonomy and four design goals through a follow-up probe (N=3).
- Designed RAVA to record agent-made design choices, anchor explanations and alternatives to the finished visualization, and turn users' retain-or-reconsider decisions into structured revision guidance.
- Conducted an expert review (N=2) and a counterbalanced within-subjects study (N=12); RAVA elicited more reflective explanations and targeted revisions without materially increasing task time or cognitive load.

Mixed-Initiative AI for Narrative Writing [UIST '26, accepted] Nov 2025 – Apr 2026
Project Lead, First Author, DVA Lab, SUSTech | Advisor: Prof. Yuxin Ma *Shenzhen, China*

- Led the project from idea to publication; conducted a formative study with nine fiction writers that identified IOED-driven latent knowledge gaps as a key barrier in unfamiliar-domain writing, motivating a mixed-initiative approach that surfaces gaps while preserving author agency.
- Designed and implemented VeriForge's mixed-initiative workflow, combining proactive gap detection, dual-stream querying, spatial knowledge curation, and source-traceable graph-based RAG.
- Demonstrated in a within-subjects study (N=12) that VeriForge substantially improved blind-spot alerting (6.33 vs. 3.00; $p = .001$, $r = .885$) and multiple dimensions of creativity support; blind ratings by two published authors further indicated stronger domain competence and more natural integration of domain details.

Visitor Simulation for Museum Exhibition Design [UIST '26, accepted] 2025 – Present
Co-author, DVA Lab, SUSTech | Advisor: Prof. Yuxin Ma *Shenzhen, China*

- Co-shaped the core research framing, translating prior experience as a volunteer docent at the Jilin Provincial Museum into a dual-layer model that connects observable visitor behavior with cognitive, affective, and meaning-related internal responses.
- Designed the formative study (N=7) and helped translate its findings into three system design goals: diverse visitor personas, dynamic multi-granular feedback, and evidence-based design iteration.
- Mobilized a personal network across museums and cultural institutions to lead participant recruitment, securing perspectives spanning museum practice and research, including experts with up to 24 years of domain experience.

ACADEMIC SERVICE

Reviewer, ACM CHI 2027

Co-organizer, Somatic City: A Biosensing Walking Workshop Aug 7–9, 2026
DePaul University & SUSTech *Chicago, IL, USA*
Advisors: Prof. Mirna Zordan, Prof. LeAnne Wagner, and Prof. David Ramsey

- Co-organized a three-day workshop on embodied HCI and urban sensing, coordinating on-site operations, participants, logistics, and resources across DePaul University and SUSTech.
- Led biosensing walks through Chicago for collecting EEG, electrodermal activity, and heart-rate variability data, then facilitated an interdisciplinary hackathon that translated the data into design concepts and prototypes.

HONORS & AWARDS

SUSTech Global Exchange Scholarship (RMB 10,000) Sep 2026
UC Davis GREAT Program Aug 2026
Outstanding Research Award (one of 3 recipients from 22 participants)
Selected Symposium Oral Presenter (one of 5 presenters from 22 participants)

SKILLS

Technical & Design: Python, TypeScript, Vue 3, FastAPI, Neo4j, Figma, LaTeX
Research Methods: Formative Interviews, Mixed-Methods User Studies, Within-Subjects Experiments, Think-Aloud Studies, Usability Evaluation, Qualitative Coding, Statistical Analysis
Languages: English (IELTS 7.5: L 8.5, R 8.0, W 7.0, S 6.5), Mandarin Chinese (Native)